

VIET NAM NATIONAL UNIVERSITY HO CHI MINH CITY
UNIVERSITY OF SCIENCE



NHẬP MÔN DỮ LIỆU LỚN

INTRODUCTION TO BIG DATA ANALYSIS

Lecturers

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Lab 01: Introduction to Hadoop Ecosystem

1. Setup a Hadoop cluster

1.1. Requirements

Install a Hadoop cluster with a pseudo-distributed configuration, then follow the instructions to perform some of common HDFS commands.

- All steps instructed in the following sections must be screenshotted and put into a file named *Report.pdf* with sufficiently detailed explanation to prove your understanding. Even if you encounter any error during the process, you must provide information about that error, .i.e your Google Search's results and/or debugging analyses, as well as the approach taken to resolve it in your report, at associated step in which you encounter it.
- You're required to run a jar file, take the output of that file and save it to `<StudentID>_verification.txt` which will be used to verify your work.

1.2. Expected outcome

- Know how to setup a Hadoop cluster by yourself. It will be used in the next labs.
- Know how to work with file permissions on HDFS through command line interface.

1.3. Instructions

- Install Hadoop, you can choose one of the following methods to install a Hadoop cluster:
 - o For Ubuntu users: Follow the official tutorial from Apache to setup a Hadoop cluster in pseudo-distributed mode. [Link](#)
 - o Docker-based container: [Link](#)
 - o Install Ubuntu VM and install Hadoop inside it: [Link](#) (VirtualBox) or [Link](#) (VMWare)
 - o WSL: [Link](#)
- Create a folder with path `/hcmus` on HDFS.
- Create a user named `khtn_<StudentID>`.
- Create a subfolder at `/hcmus/<StudentID>` and upload a file into it.
- `Chmod 744 /hcmus/<StudentID>` and set the owner of that subfolder to the user named `khtn_<StudentID>`.
- Run the attached JAR file named **hadoop-test.jar**:

```
java -jar /path/to/jar/file.jar <YOUR_HDFS_PORT> /hcmus/<StudentID>
```

- There will be sometimes you encounter an error related to network configuration and information retrieval such as “Exception in thread main null ... Main getMac ...”.
 - This is when you should check for whether you have downloaded the latest jar file because the latest version will also print out debugging information for this kind of errors. As the last version of the above jar file is accompanied with this version of the problem statement which contains these lines of instruction, it is highly unlikely that you will read these lines while getting no information from the jar.
 - Error details: this error means that the Java Virtual Machine (JVM) used to execute the above jar file is having issue with identifying network configuration from your installed environment which is an issue specifically comes from your setup process and/or your host environment. The latest version of the jar file will not fix it but it helps you identify the detailed source of this error.
 - After you have the debugging information printed from the latest jar file, you may read it and research to identify the main cause buried under your Hadoop environment, then try to fix it in an appropriate way. All this process (if you encounter the error) is expected to be included within the report.
- Find the file named `<StudentID>_verification.txt` which is generated after running the JAR file and include it in your submission.

2. Warm up with Word Count

2.1. Requirements

Write a program to count the number of words starting with letters a, f, j, g, h, c, m, u, s (case insensitive) in the attached dictionary file named **words.txt**. You are expected to use one programming language among the following three: Java, Scala, and Python.

Then, export a TSV-formatted file for submission where each row contains a letter in the above list and the number of words starting with it. An invalid submission will render a score of 0 (zero) for this labwork, so be careful. An example of the submitted TSV file would be:

a	1
f	3
j	5

2.2. Expected output

Get familiar with MapReduce through the most basic problem: Word Count.

3. Submission Guideline

This lab requires a group's submission where the work of your group's members is compressed into a single file and only one representative may submit this file on Moodle. The submission file contains a single folder named *<FirstMemberID>* where student ID of the first member (that your group has registered in earlier form) is used. This folder's internal structure contains the members' results with each member's associated folder is named as *<StudentID>* based on the member's student ID. Specifically the structure will be as follow

```
<FirstMemberID>
|--- <StudentID>
|   |--- docs
|   |   |--- Report.pdf
|   |   |--- <StudentID>_verfication.txt
|   |   |   ...
|   |--- src
|   |   |--- WordCount
|   |   |--- | results.txt
|   |   |--- |   ...
|   |--- README (optional for instructions on how to run code)
|   ...
```

You must strictly follow the above file structure and compress the whole folder into a ZIP file named *<FirstMemberID>.zip*, which is your final file to be submitted to Moodle.

- Ensure your code is well-documented with clear comments.
- Include all necessary files, logs, and screenshots to verify successful execution.

Happy Coding and Best of Luck!

The Lab Instructor./.